

CG3.401 INCM - Assignment 1

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By turning in this assignment, I agree to the IIIT-H honor code and declare that all of this is my own work.

1 Passive Membrane Model

Equation

$$\frac{du(t)}{dt} = -\frac{u(t) - u_{\text{rest}}}{RC} + \frac{1}{C}I(t)$$

where,

$u(t)$	=	membrane potential at time t
u_{rest}	=	resting potential
R	=	membrane resistance
C	=	membrane capacitance
$I(t)$	=	external input current

Simulation

Euler's Integration Equation

$$u(t + \Delta t) = u(t) + \Delta t \left[-\frac{u(t) - u_{\text{rest}}}{RC} + \frac{1}{C}I(t) \right]$$

where,

$$\Delta t = \text{discrete time step for Euler Integration}$$

Figures

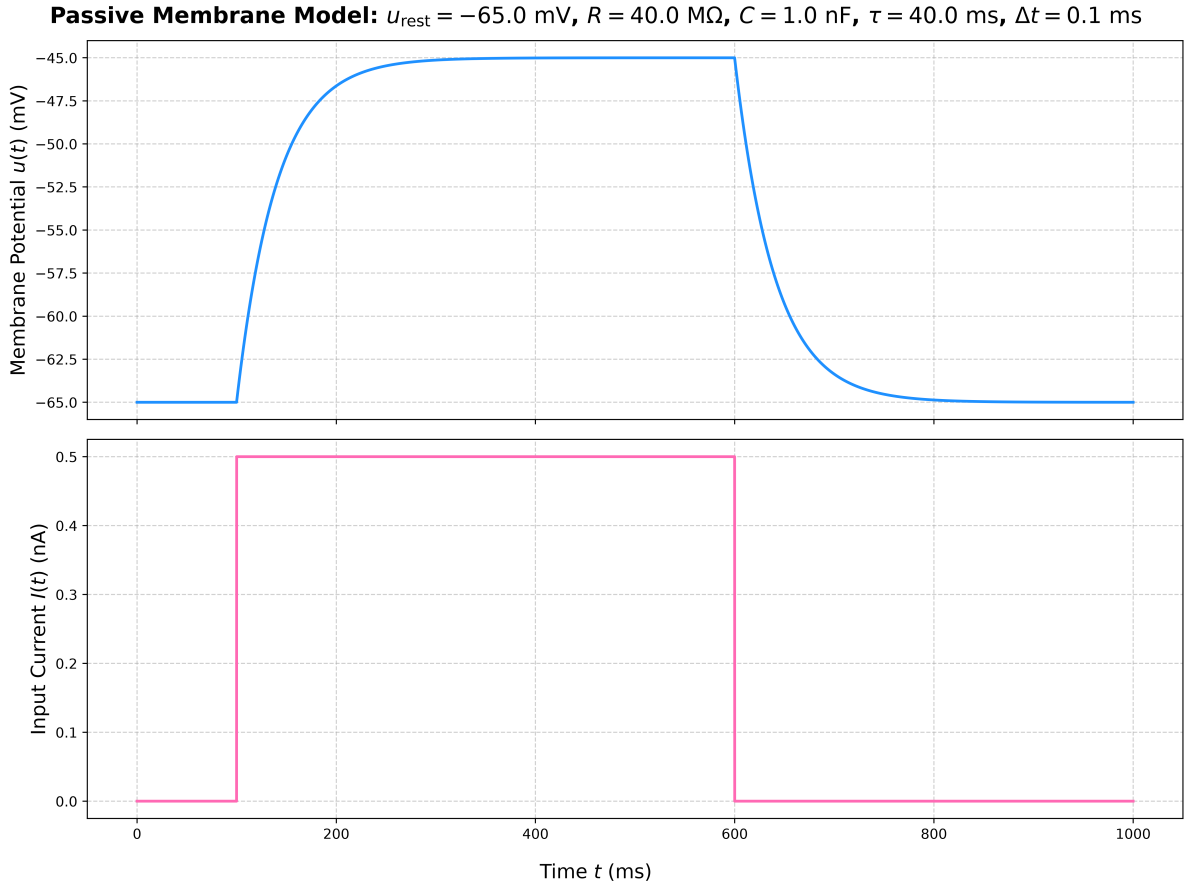


Figure 1: Passive Model Membrane Simulation

Comments

- The input current is a step function that starts at $t = 100 \text{ ms}$ and ends at $t = 600 \text{ ms}$ with a steady current value of 0.5 nA during this interval.
- The membrane potential increases exponentially under the effect of the input current until the current persists.
- The membrane potential decreases exponentially when the external current ceases to exist.
- The maximum membrane potential achieved is close to -45 mV .

Theoretical Results

When the constant current $I_0 = 0.5 \text{ nA} = 5 \times 10^{-10} \text{ A}$ is applied, the asymptotic steady-state potential reached as $t \rightarrow \infty$ is:

$$u_{\infty} = u_{\text{rest}} + R \cdot I_0 = -65 \text{ mV} + (40 \text{ M}\Omega)(0.5 \text{ nA}) = -65 \text{ mV} + 20 \text{ mV} = \mathbf{-45 \text{ mV}}$$

2 Leaky Integrate and Fire Model

Equation

$$\tau \frac{du(t)}{dt} = -(u(t) - u_{rest}) + \frac{I(t)}{g_L}$$

where,

- $u(t)$ = membrane potential at time t
- u_{rest} = resting potential
- τ = membrane time constant, ($\tau = RC$, where R and C are membrane resistance and capacitance respectively)
- g_L = leak conductance, ($g_L = \frac{1}{R}$, where R is the membrane resistance)
- $I(t)$ = external input current

The external input current is given by:

$$I(t) = I_0 \sin(2\pi ft)$$

where,

$$I_0 = 0.25nA$$

Simulation

Using Euler's Integration with time step Δt we have,

$$u(t + \Delta t) = u(t) + \Delta t \left[-\frac{u(t) - u_{rest}}{\tau} + \frac{1}{\tau g_L} I(t) \right]$$

Approximating membrane potential

For each input frequency, we simulate the neuron for a sufficiently long time to observe the initial transient followed by at least five complete cycles of the steady-state response. We discard at least the first 5τ of the simulation before estimating the amplitude and phase.

After the transient has decayed, the membrane potential can be approximated as :

$$u(t) = u_0 + A \sin(2\pi ft + \phi)$$

I used least squares linear regression method for estimating A and ϕ from the simulated observations.

Figures

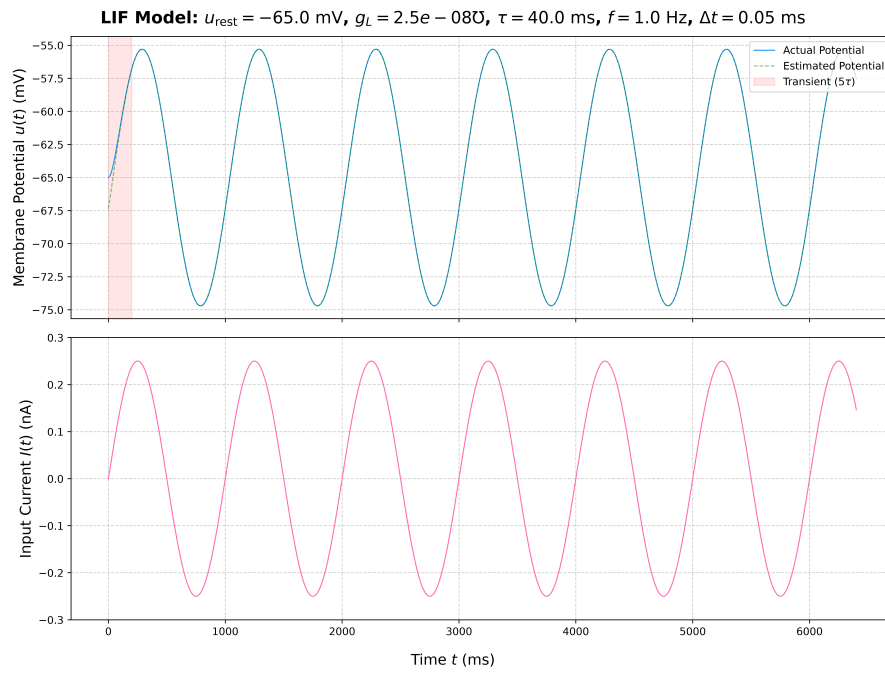


Figure 2: [Top] Actual membrane potential with an initial transient along with the estimated potential. [Bottom] Sinusoidal input current with frequency 1 Hz.

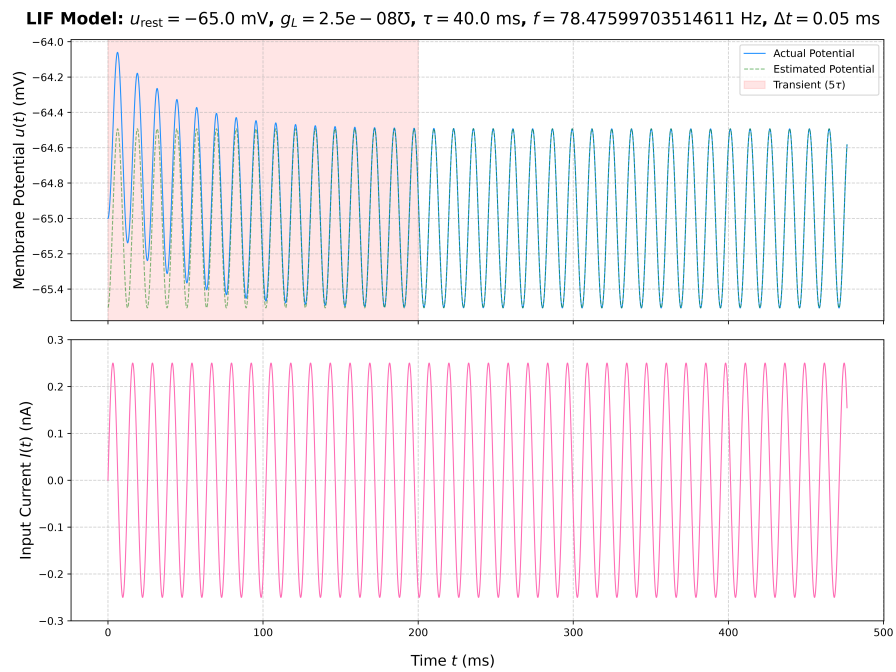


Figure 3: [Top] Actual membrane potential with an initial transient along with the estimated potential. [Bottom] Sinusoidal input current with frequency 78.47599 Hz.

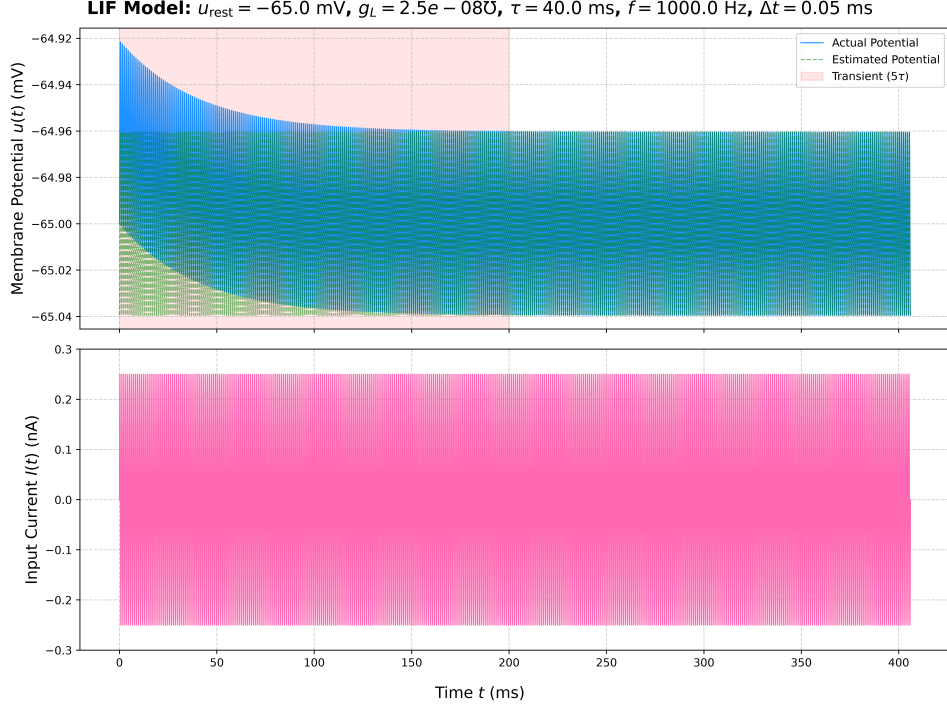


Figure 4: [Top] Actual membrane potential with an initial transient along with the estimated potential. [Bottom] Sinusoidal input current with frequency 1000 Hz.

NOTE: The figures for other frequencies between 1 Hz and 1000 Hz are included in the zip file under folder q2.

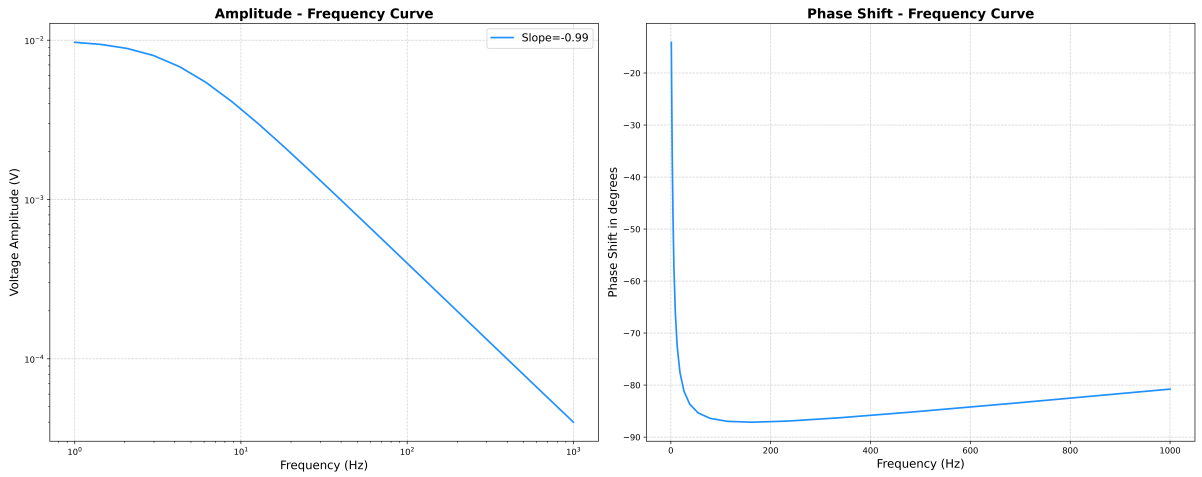


Figure 5: [Left] Amplitude vs Frequency curve on log-log scale of the steady-state amplitude of the membrane potential oscillation for frequencies between 1 Hz and 1000 Hz. [Right] Phase shift of the membrane-potential oscillation relative to the input current over the same range of frequencies.

Comments

- At low frequencies ($f \lesssim 4$ Hz), the amplitude response is flat and at its maximum ($A \approx 10$ mV = $R \cdot I_0$).
- At high frequencies ($f \gtrsim 10$ Hz), the amplitude steadily decreases with an approximate downward slope of -1 . High-frequency amplitude decays inversely with frequency: $A \propto f^{-1}$.
- This means that the membrane behaves as a Low-Pass Filter. It transmits slow injected currents into voltage changes, but attenuates rapid fluctuations.
- Why low frequencies pass?
When the current oscillates slowly, the membrane capacitance C has ample time to charge and discharge to its steady-state potential ($\Delta u = R \cdot I_0 = 40 \text{ M}\Omega \times 0.25 \text{ nA} = 10 \text{ mV}$).
- Why high frequencies are blocked?
When oscillations are very fast, the capacitive reactance $X_C = \frac{1}{2\pi f C}$ becomes very small. High-frequency current is shorted through the capacitor before it can build up a significant trans-membrane voltage.
- As frequency increases, the peak of the membrane potential lags behind the peak of the input current:
- At $f \rightarrow 0$: $\phi \approx 0^\circ$ (voltage is in phase with current).
- At $f \gg 0$: $\phi \rightarrow -90^\circ$ ($-\pi/2$ rad, pure capacitive delay).

3 Advanced Leaky Integrate-and-Fire (LIF) Neuron Model

Equation

Extends the subthreshold LIF model by introducing a firing threshold, voltage reset, and refractory period. Whenever the membrane potential reaches the firing threshold u_{th} , a spike is recorded. Immediately after a spike, the membrane potential is reset to u_{reset} and held there for the duration of the refractory period t_{ref} .

3.1 LIF Simulation Function

```
1 def simulate_adv_lif(u_rest, u_th, u_reset, gL, tau, I_t, t_ref,
2   delta_t=1e-4, t_0=0, t_1=1):
3     t = t_0
4     u_t = u_rest
5
6     # Store results
7     U = []
```

```

8   I = []
9   T = []
10  spikes = []
11
12  while t < t_1:
13      i_t = I_t(t) # Get input current value
14      u_t = u_t + delta_t * (
15          -(u_t - u_rest) / (tau) + i_t / (tau * gL)
16      ) # Euler Integration
17
18      U.append(u_t)
19      I.append(i_t)
20      T.append(t)
21
22      if u_t >= u_th: # Spike generated
23          spikes.append(True)
24
25          t_r = 0
26          while t_r < t_ref: # Refractory period
27              t += delta_t
28              i_t = I_t(t)
29              U.append(u_reset)
30              I.append(i_t)
31              T.append(t)
32              t_r += delta_t
33              spikes.append(False)
34
35          u_t = u_reset
36
37      else:
38          spikes.append(False)
39
40      t += delta_t
41
42  return np.array(U), np.array(I), np.array(T), np.array(spikes)
43

```

Listing 1: Python Implementation of LIF Simulation Model

3.2 Constant Current and Rheobase

Drive the neuron with a constant input current $I(t) = I_0$. Test constant currents between 50 pA and 300 pA in steps of 15 pA.

Figures

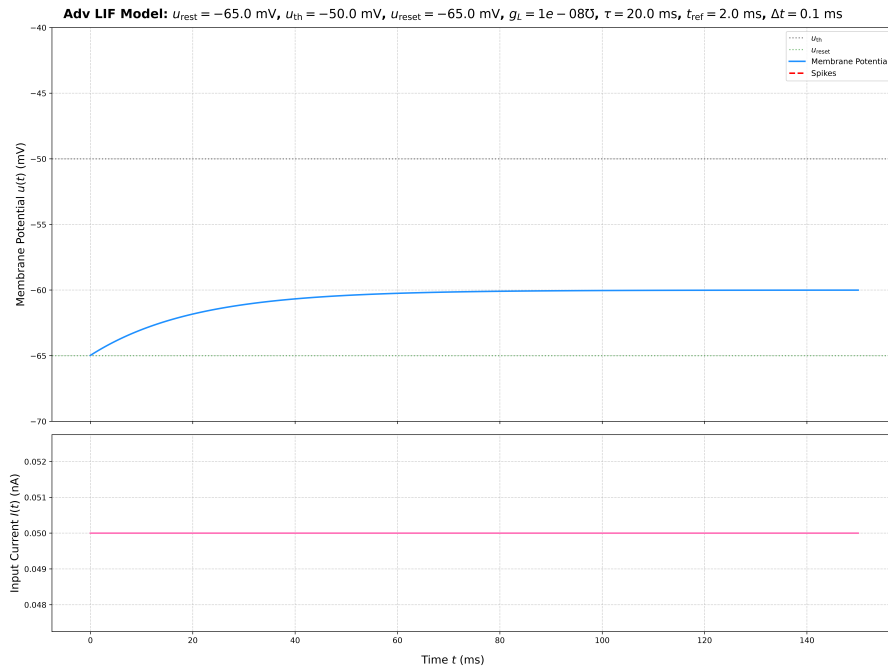


Figure 6: [Top] Membrane potential stays below threshold and reaches -60 mV. [Bottom] Sub-threshold input current of 50 pA.

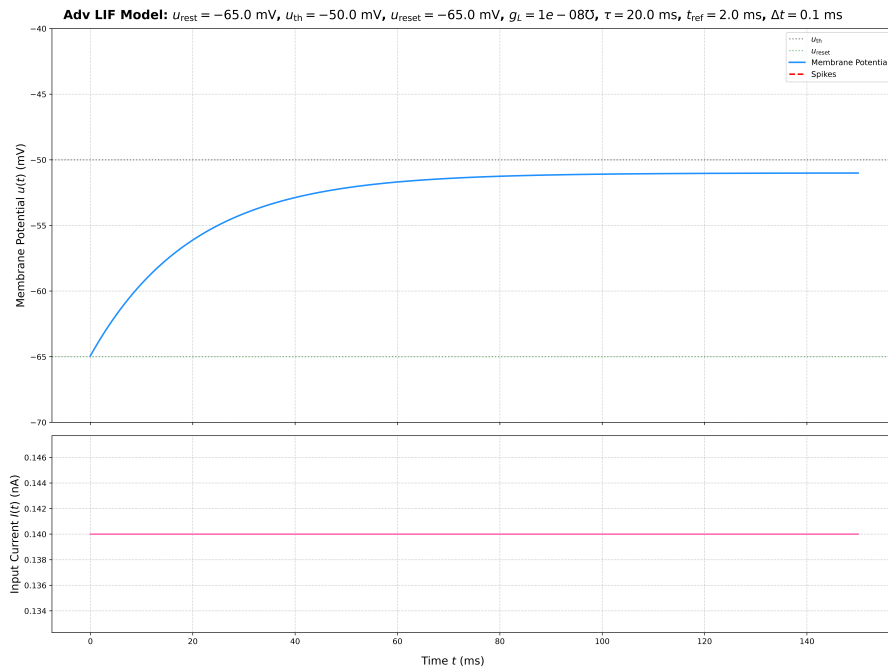


Figure 7: [Top] Membrane-potential just stays below the threshold of -50 mV. [Bottom] Sub-threshold input current of 140 pA.

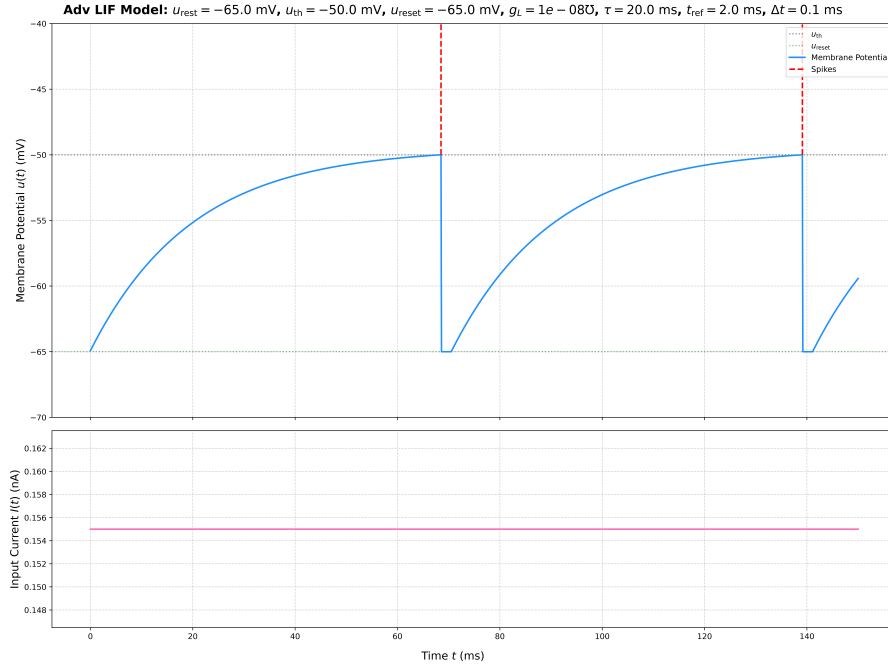


Figure 8: [Top] Membrane potential goes above the threshold of -50 mV and spikes. [Bottom] Supra-threshold input current of 155 pA.

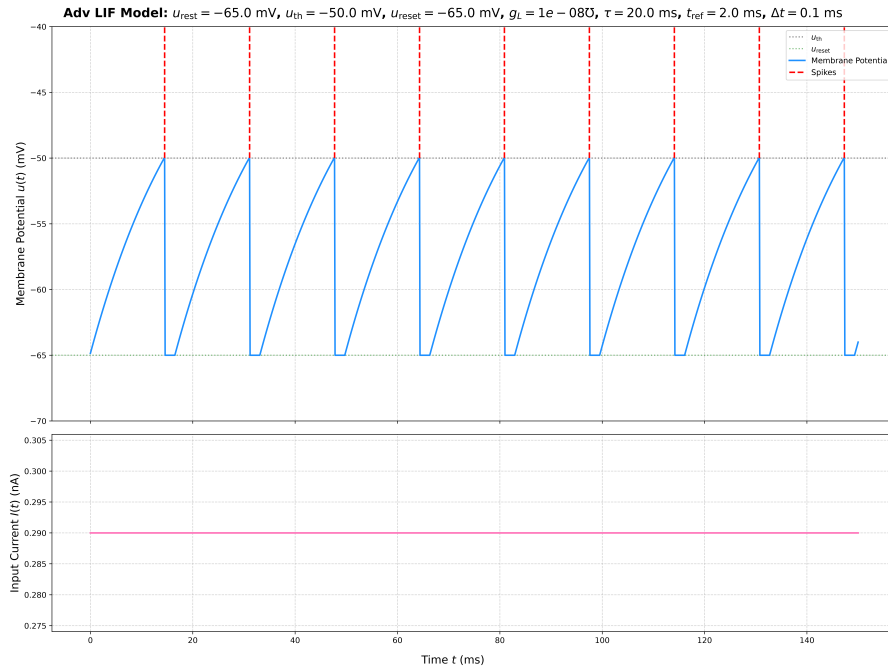


Figure 9: [Top] Membrane potential goes above the threshold of -50 mV and spikes. [Bottom] Supra-threshold input current of 290 pA.

Comments

- The smallest current that produced repeated firing from our tested values of input current is 155 pA. Hence, estimated rheobase current = 155 pA.
- The theoretical rheobase current is given by: $I_{\text{rh}} = g_L(u_{\text{th}} - u_{\text{rest}}) = 10 \text{ nS} \times (-50 \text{ mV} - (-65 \text{ mV})) = 150 \text{ pA}$.
- The numerical estimate (155 pA) differs slightly from the theoretical value (150 pA) due to the following factors:

Factor	Mechanism / Cause	Effect on Numerical Estimate
Current Step Size ($\Delta I = 15 \text{ pA}$)	Coarse grid discretization leading to bigger intervals	Introduces discretization uncertainty ($155 \pm 15 \text{ pA}$); the true threshold 150 pA falls between test values.
Finite Time Horizon (T_{sim})	Diverging spike latency ($t_{\text{spike}} \rightarrow \infty$ as $I \rightarrow I_{\text{rh}}^+$)	Currents slightly above 150 pA require long charging times ($t \gg \tau$) to reach threshold and do not fire within a finite T_{sim} .
Integration Step Size ($\Delta t = 0.1 \text{ ms}$)	Forward Euler local truncation error $\mathcal{O}(\Delta t^2)$	Discretization error slightly delays the rate of voltage depolarization near the threshold.

Table 1: Factors leading to difference in the theoretical and estimated values of I_{rh} .

3.3 Effect of the Membrane Time Constant

The membrane time constant τ determines how rapidly the membrane potential responds to an input. Keeping all other model parameters fixed, including the leak conductance g_L , and using a constant input current of $I(t) = 200 \text{ pA}$, we repeat the simulation using the following membrane time constants: $\tau = 10 \text{ ms}$, $\tau = 20 \text{ ms}$, $\tau = 40 \text{ ms}$.

Figures

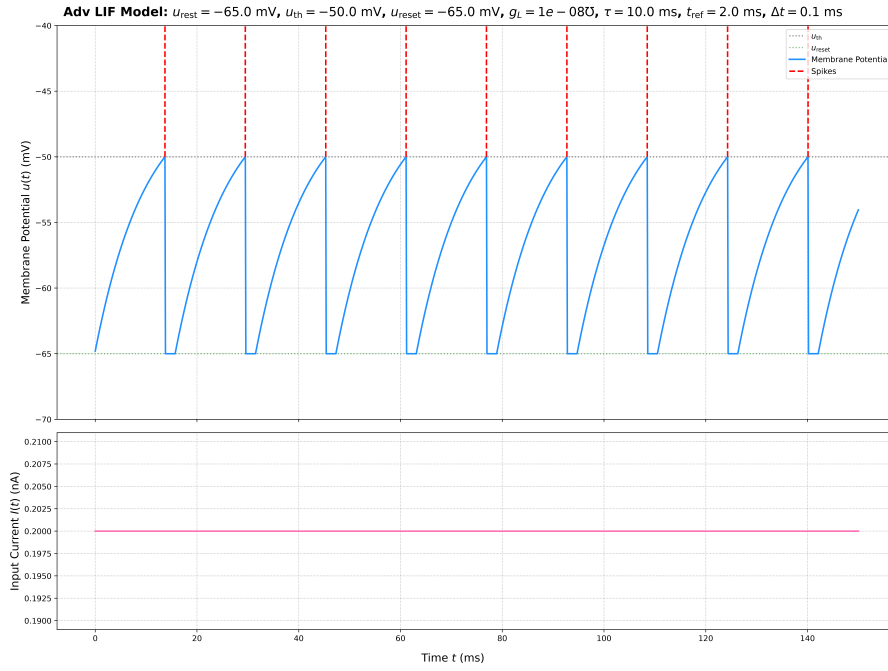


Figure 10: [Top] Membrane potential trace with $\tau = 10$ ms. $T_{ISI} = 15.8$ ms, $f = 63.29$ Hz. [Bottom] Constant input current of 200 pA.

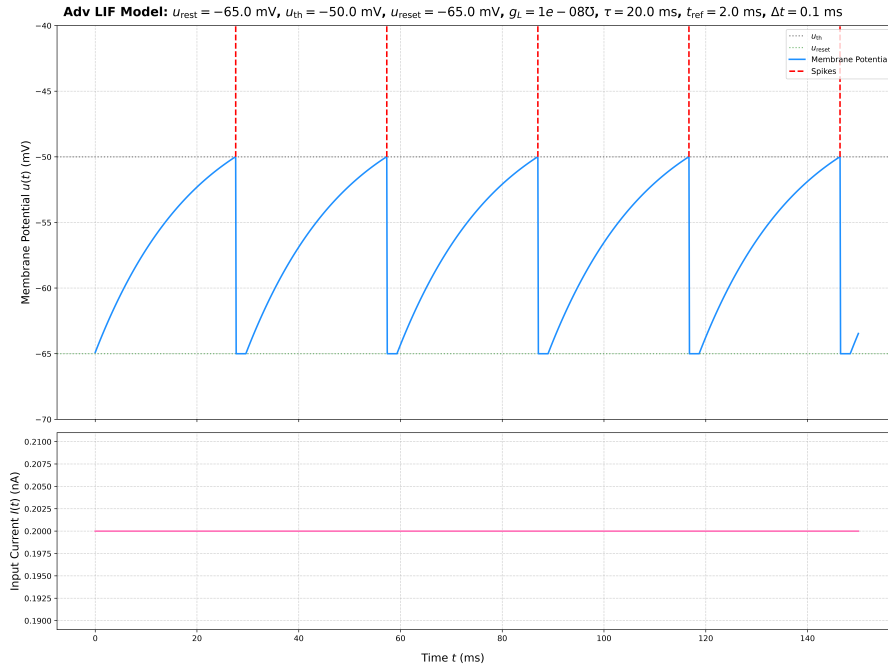


Figure 11: [Top] Membrane potential trace with $\tau = 20$ ms. $T_{ISI} = 29.7$ ms, $f = 33.67$ Hz. [Bottom] Constant input current of 200 pA.

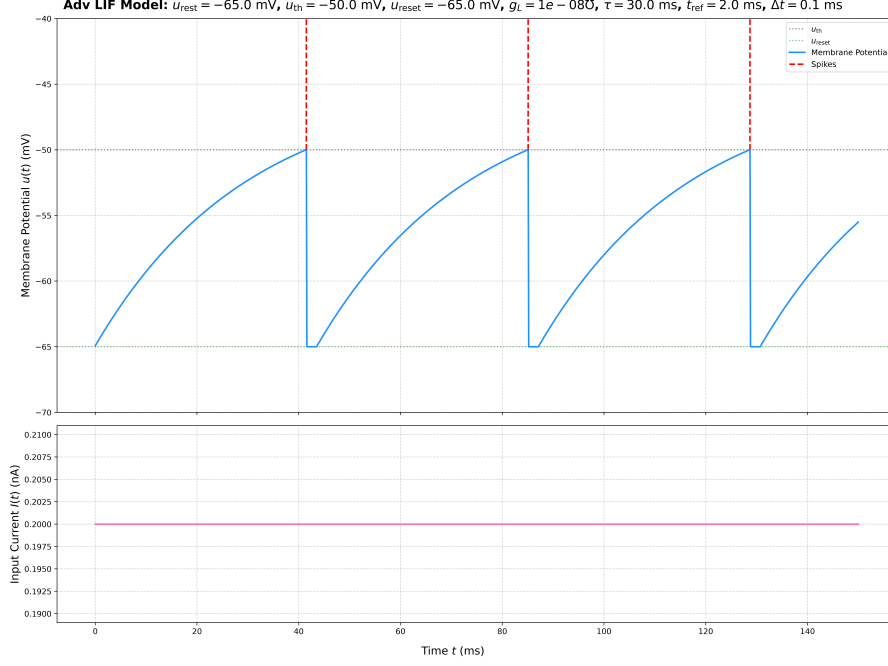


Figure 12: [Top] Membrane potential trace with $\tau = 30\text{ms}$. $T_{ISI} = 43.6\text{ms}$, $f = 22.94\text{Hz}$. [Bottom] Constant input current of 200 pA.

Comments

- With conductance g_L held constant, the time constant is directly proportional to membrane capacitance: $\tau = \frac{C}{g_L}$ (or $\tau = RC$).
- A smaller capacitor stores less electrical charge ($Q = C\Delta u$) and charges up much faster for a given injected current ($\frac{du}{dt} = \frac{I_0 - g_L(u - u_{\text{rest}})}{C}$). Consequently, the membrane potential depolarizes faster and reaches the firing threshold u_{th} in less time.
- A larger capacitor requires more accumulated charge to produce the same voltage change, resulting in a slower voltage rise towards threshold. ($15.8 < 29.7 < 43.6$).
- The inter-spike interval is given by:

$$T_{\text{ISI}} = t_{\text{ref}} + \tau \ln \left(\frac{u_{\infty} - u_{\text{reset}}}{u_{\infty} - u_{\text{th}}} \right) = t_{\text{ref}} + \frac{C}{g_L} \ln \left(\frac{u_{\text{rest}} + \frac{I_0}{g_L} - u_{\text{reset}}}{u_{\text{rest}} + \frac{I_0}{g_L} - u_{\text{th}}} \right)$$

- The firing rate is $f = \frac{1}{T_{\text{ISI}}}$.
- Decreasing τ (Lower C): Reduces the charging time $T_{\text{charge}} \propto C$, which shortens T_{ISI} and increases the firing frequency f .
- Increasing τ (Higher C): Lengthens the charging phase, increasing T_{ISI} and decreasing the firing frequency f . ($63.29 > 33.67 > 22.94$).
- At very high currents ($I_0 \rightarrow \infty$), the charging time approaches zero and the firing rate saturates at the refractory limit $f_{\text{max}} = \frac{1}{t_{\text{ref}}}$.

- Changing τ (or C) does NOT alter the steady-state membrane potential.
- In the absence of a firing threshold, at steady-state ($t \rightarrow \infty$), the capacitive current ceases entirely ($\frac{du}{dt} = 0 \implies I_{\text{cap}} = C \frac{du}{dt} = 0$).
- All injected current flows purely through the leak conductance:

$$0 = -g_L(u_\infty - u_{\text{rest}}) + I_0 \implies u_\infty = u_{\text{rest}} + \frac{I_0}{g_L}$$

- Capacitance C (and hence $\tau = C/g_L$) only controls the rate of charging, not the steady-state DC voltage, which depends exclusively on u_{rest} , g_L , and I_0 .

3.4 Input–Output Relationship

How the firing rate of the neuron depends on the strength of the input current by varying the constant input current from 50 pA to 300 pA.

Figures

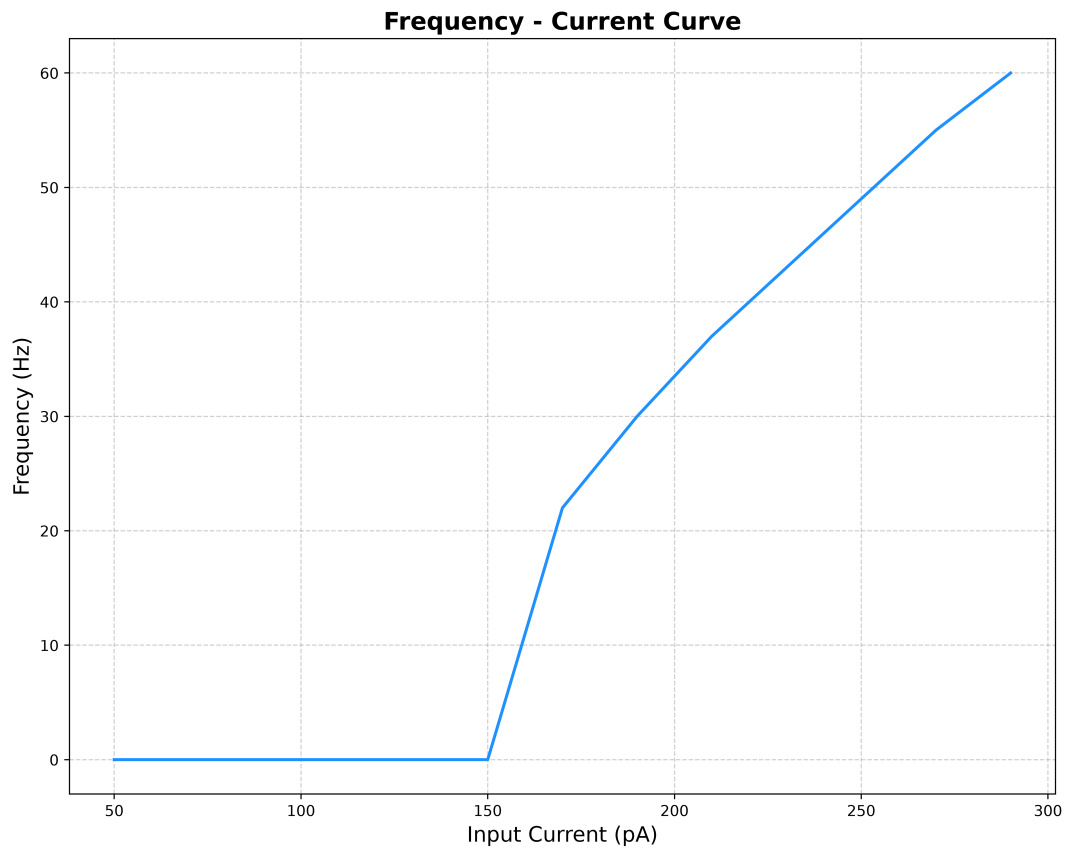


Figure 13: F-I Curve (firing rate in Hz versus input current in pA)

Comments

The relationship between input current I_0 and firing rate $f(I_0)$ (the F-I curve) exhibits three distinct regimes:

1. Below Rheobase ($I_0 < I_{\text{rh}}$) — Subthreshold Regime
 - Firing Rate: $f = 0$ Hz.
 - The steady-state potential $u_\infty = u_{\text{rest}} + \frac{I_0}{g_L}$ remains below the threshold u_{th} .
 - No action potentials are generated.
2. Just Above Rheobase ($I_0 \gtrsim I_{\text{rh}}$) — Onset of Firing
 - Firing Rate: Starts from 0 Hz and rises steeply.
 - As $I_0 \rightarrow I_{\text{rh}}^+$, the latency to spike diverges ($T_{\text{ISI}} \rightarrow \infty$), resulting in arbitrarily low firing frequencies ($f \rightarrow 0^+$), characteristic of a Saddle-Node bifurcation.
3. Moderate to High Currents ($I_0 \gg I_{\text{rh}}$) — Growth & Saturation
 - Moderate Currents: The charging time decreases rapidly with current, leading to a increase in firing rate with input current.
 - Very High Currents ($I_0 \rightarrow \infty$):
 - The charging time becomes negligible ($T_{\text{charge}} \rightarrow 0$) compared to the absolute refractory period t_{ref} .
 - The inter-spike interval asymptotically approaches $T_{\text{ISI}} \approx t_{\text{ref}}$.
 - The firing rate saturates at a maximum upper limit:

$$f_{\text{max}} = \frac{1}{t_{\text{ref}}} = \frac{1}{2 \text{ ms}} = 500 \text{ Hz}$$

3.5 Effect of Noise on Spike Timing

To investigate the effect of fluctuations, we use an input current of the form

$$I_n = I_{\text{mean}} + \sigma \zeta_n$$

where,

$$\zeta_n \sim \mathcal{N}(0, 1)$$

Figures

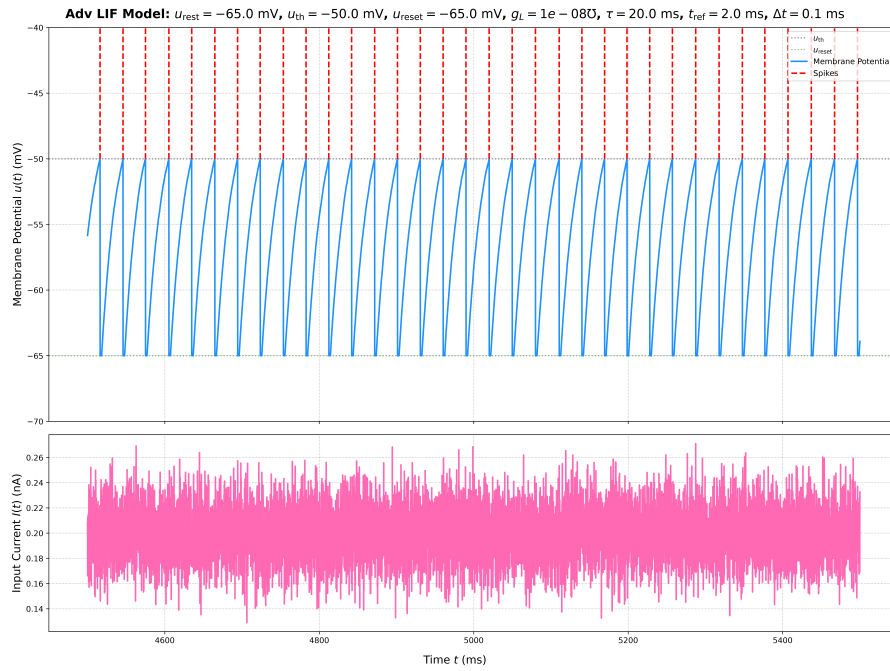


Figure 14: [TOP] Low Noise Membrane potential trace. [BOTTOM] Input current with $I_{mean} = 200pA$ with $\sigma_{low} = 20pA$.

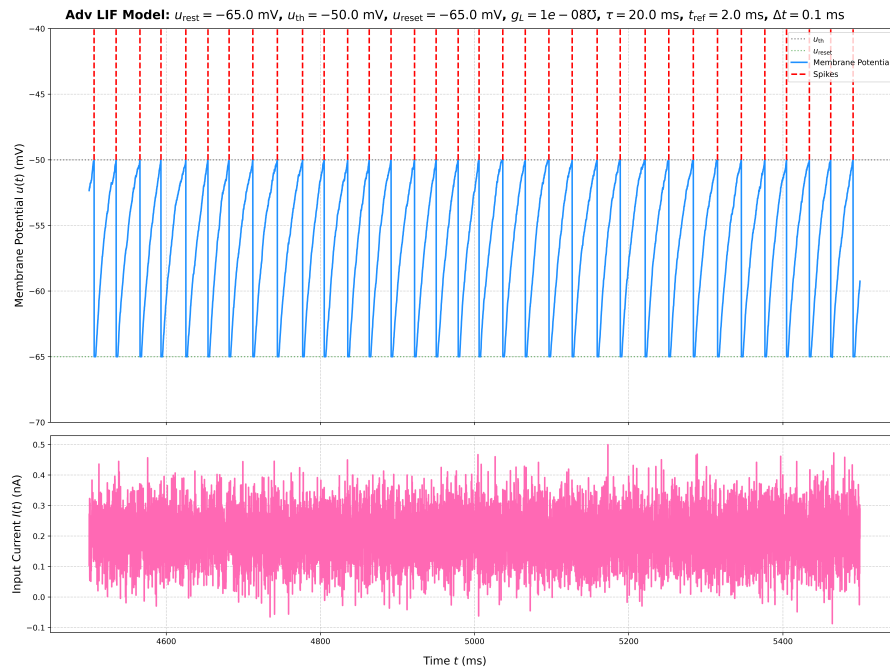


Figure 15: [TOP] High Noise Membrane potential trace. [BOTTOM] Input current with $I_{mean} = 200pA$ with $\sigma_{high} = 80pA$.

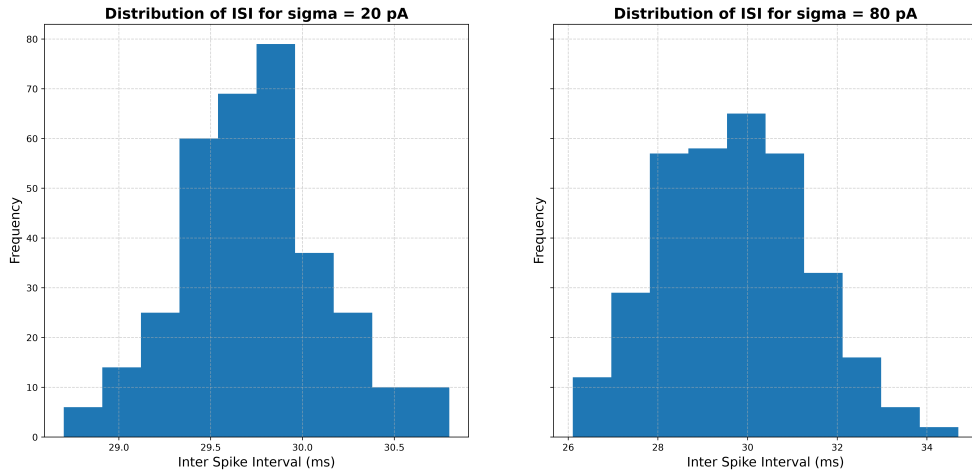


Figure 16: [LEFT] Histogram of Inter-Spike Intervals (ISI) for low noise scenario. $CV = 0.0127$. [RIGHT] Histogram of ISI for high noise scenario. $CV = 0.0541$.

Comments

- Broadening of the ISI Distribution (Variance Increases):
 - $\sigma = 20$ pA (Low Noise): The ISI distribution forms a narrow Gaussian-like peak centered near the deterministic ISI, indicating relatively regular firing.
 - $\sigma = 80$ pA (High Noise): The ISI distribution spreads out (broad variance), developing positive skewness with a longer tail towards both shorter and longer intervals.
- Increase in Coefficient of Variation (CV)
 - $CV \approx 0$: Highly regular / deterministic firing.
 - $CV \rightarrow 1$: Highly irregular / memoryless firing (resembles a Poisson process).
 - CV increases monotonically with noise amplitude σ , demonstrating that stronger noise drives the neuron from a deterministic rhythm towards stochastic Poisson-like spiking.
- Even when mean current is strictly subthreshold ($I_0 < I_{\text{rheo}}$), large positive noise fluctuations can randomly drive $u(t) \geq u_{\text{th}}$, producing noise-driven / fluctuation-driven action potentials that would never occur deterministically.

NOTE: The report contains only the relevant plots required for answering the assignment questions. Remaining plots and simulation data can be found in the zip file submitted along with this report.